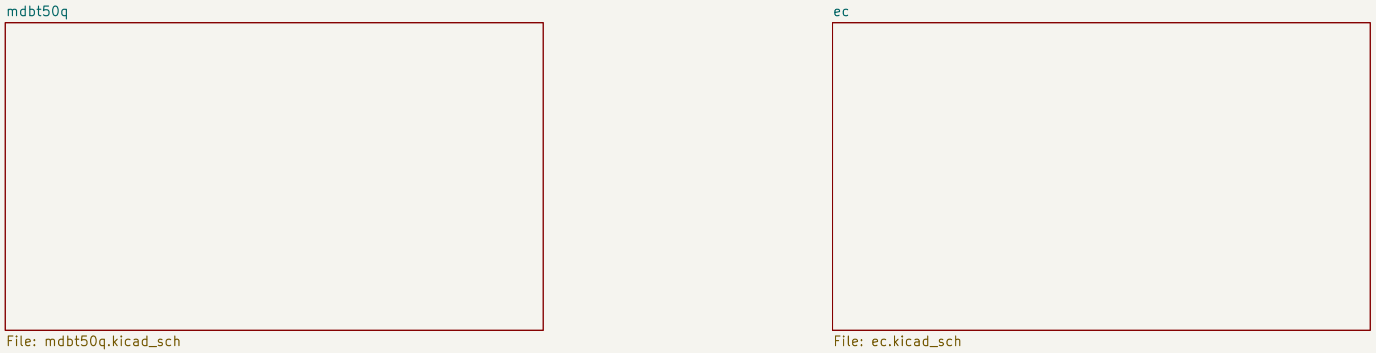
TC52 wireless: MDBT50Q-1MV2 (nRF52840) + 1S LiPo version of the wired TC52 capacitive keyboard.
Sheet mdbt50q: module, charger, USB, SWD, reset, encoder. Sheet ec: 52 electrodes, MUX x2, analog front end.
The sheets are joined by global labels: ROW1..4_DRV, INHA/S1A..S3A, INHB/S1B..S3B, SENSE (purge, P1.02),
ADC (AIN4 = P0.28), AFE_EN (P0.13), ENC_A / ENC_B / ENC_SW, +3V3 (module VDD = REG0 output), GND.
Pin table and firmware notes: docs/tc52w-build-log.md



Generated by pcb/tc52w-build/gen_sch.py

Sheet: /
File: ts52k.kicad_sch

Title: TC52 wireless

Size: A4 | Date: | **Rev: 0.1**
KiCad E.D.A. 10.0.8 | Id: 1/3

MDBT50Q−1MV2 (nRF52840). Pads 3..14 are ’standard drive, low frequency I/O only’ (Raytac spec 2.6): they carry the
encoder and the MUX select / inhibit lines (change once per column, well under 10 kHz) and the ADC input (AIN4).
Row strobes are on unrestricted pads; PURGE (P1.02) only ever sees the millivolt−level SENSE node. P0.15 (boot loader LED), NFC pins and P0.26/P0.27 (next to XL1/XL2) are unused.

32.768 kHz crystal (CL 9 pF), 12 pF x2

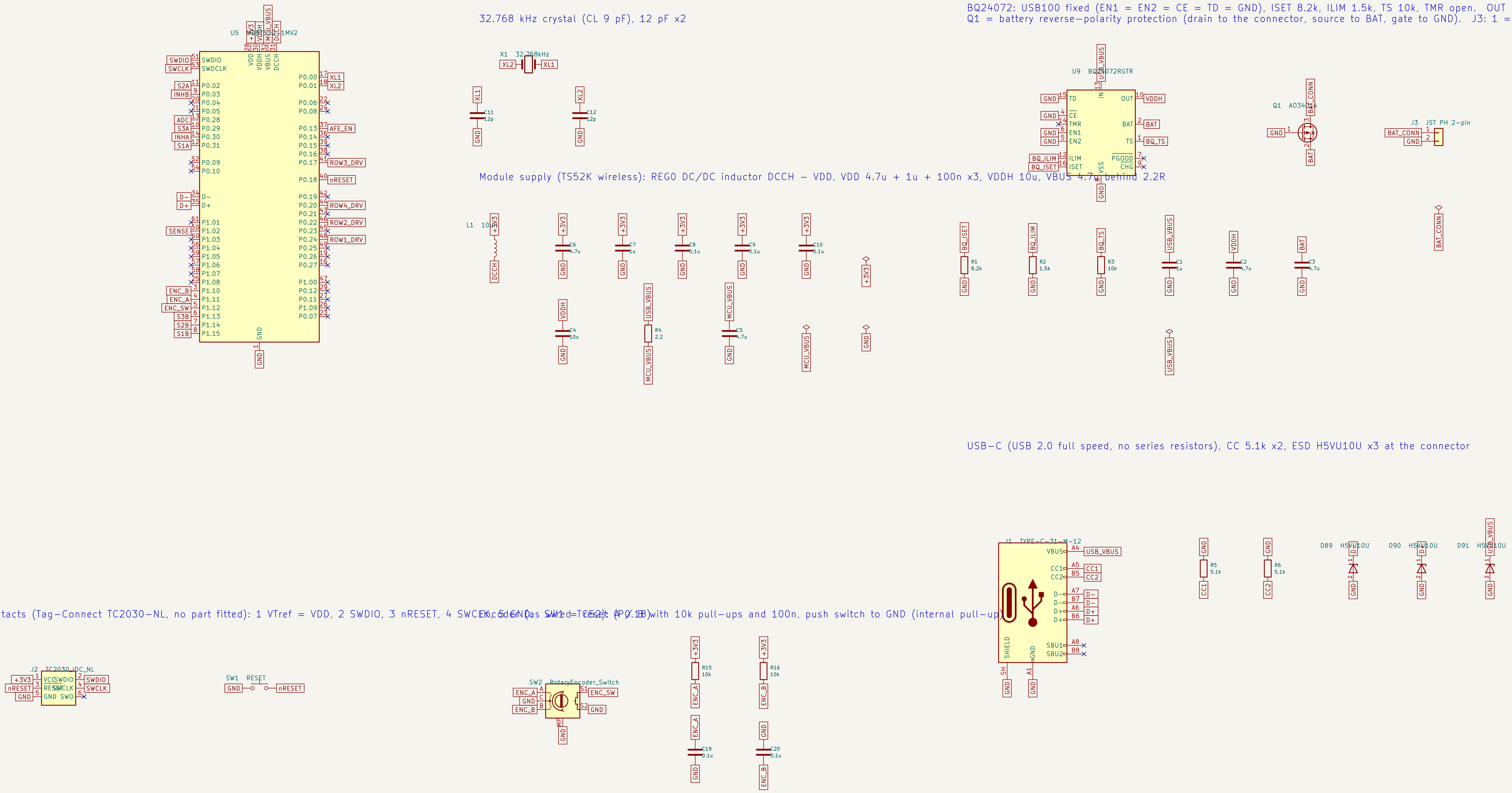
BQ24072: USB100 fixed (EN1 = EN2 = CE = TD = GND), ISET 8.2k, ILIM 1.5k, TS 10k, TMR open. OUT −> VDDH.
Q1 = battery reverse−polarity protection (drain to the connector, source to BAT, gate to GND). J3: 1 = BAT+, 2 = GND

Module supply (TS52K wireless): REG0 DC/DC inductor DCCH − VDD, VDD 4.7u + 1u + 100n x3, VDDH 10u, VBUS 4.7u behind 2.2R

USB−C (USB 2.0 full speed, no series resistors), CC 5.1k x2, ESD H5VU10U x3 at the connector

SWD contacts (Tag−Connect TC2030−NL, no part fitted): 1 VTref = VDD, 2 SWDIO, 3 nRESET, 4 SWCLK, 5 GND. SW1 = reset (P0.18)

Encoder (as wired in TC52): A/B with 10k pull−ups and 100n, push switch to GND (internal pull−up)


U5 MDBT50Q-1MV2
SWDIO
SWCLK
P0.00 XL1
P0.01 XL2
P0.13 AFE_EN
P0.17 ROW3_DRV
P0.18 nRESET
P0.20 ROW4_DRV
P0.22 ROW2_DRV
P0.24 ROW1_DRV
S2A
INHB
ADC
S3A
INHA
S1A
D−
D+
SENSE
ENC_B
ENC_A
ENC_SW
S3B
S2B
S1B
+3V3
VDDH
MCU_VBUS
DCCH
X1 32.768kHz
C11 12p
C12 12p
L1 10uH
C6 4.7u
C7 1u
C8 0.1u
C9 0.1u
C10 0.1u
C4 10u
R4 2.2
C5 4.7u
U9 BQ24072RGTR
TD
CE
TMR
EN1
EN2
ILIM
ISET
OUT
BAT
TS
PGOOD
CHG
VSS
IN
USB_VBUS
BQ_ILIM
BQ_ISET
BQ_TS
Q1 AO3401A
BAT_CONN
J3 JST PH 2-pin
R1 8.2k
R2 1.5k
R3 10k
C1 1u
C2 4.7u
C3 4.7u
J1 TYPE-C-31-M-12
VBUS
CC1
CC2
SBU1
SBU2
SHIELD
GND
R5 5.1k
R6 5.1k
D89 H5VU10U
D90 H5VU10U
D91 H5VU10U
J2 TC2030_IDC_NL
SW1 RESET
SW2 RotaryEncoder_Switch
R15 10k
R16 10k
C19 0.1u
C20 0.1u

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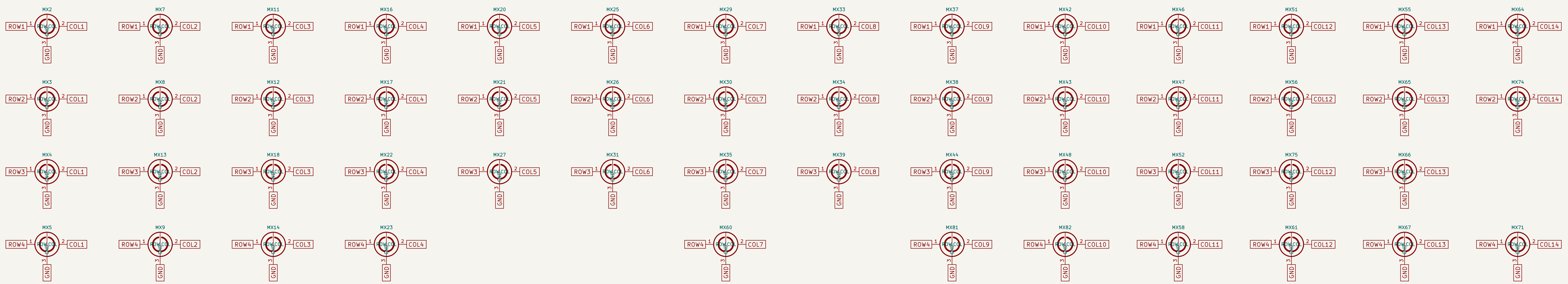
Sheet: /mdbt50q/
File: mdbt50q.kicad_sch

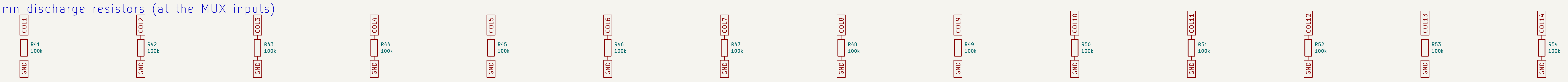
Title: TC52 wireless: MDBT50Q, power, USB, SWD, encoder

Size: A2
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 2/3

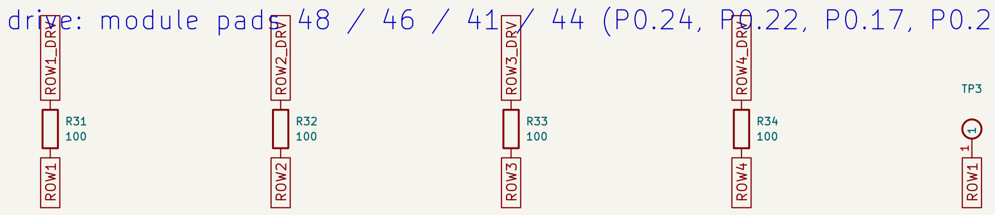
Key electrodes: 14 columns x 4 rows (design-spec 3). Pad 1 = drive (ROW), pad 2 = sense (COL), pad 3 = guard ring


MX2
MX7
MX11
MX16
MX20
MX25
MX29
MX33
MX37
MX42
MX46
MX51
MX55
MX64
MX3
MX8
MX12
MX17
MX21
MX26
MX30
MX34
MX38
MX43
MX47
MX56
MX65
MX74
MX4
MX13
MX18
MX22
MX27
MX31
MX35
MX39
MX44
MX48
MX52
MX75
MX66
MX5
MX9
MX14
MX23
MX60
MX81
MX82
MX58
MX61
MX67
MX71
ROW1
ROW2
ROW3
ROW4
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
COL9
COL10
COL11
COL12
COL13
COL14
GND

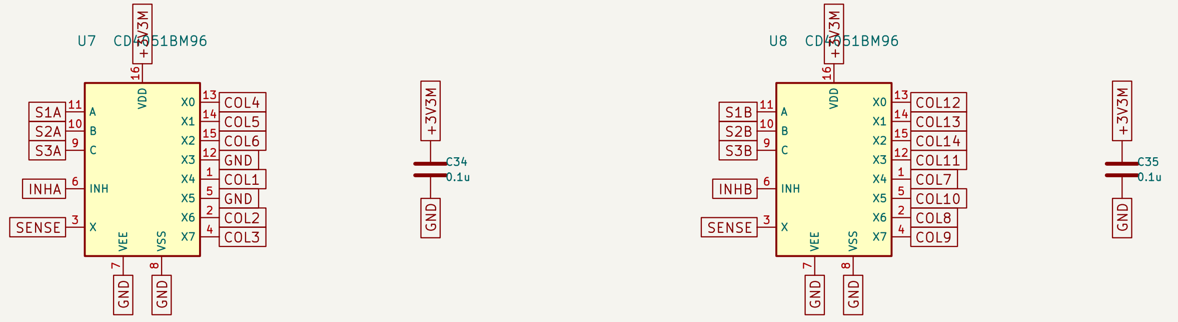
Column discharge resistors (at the MUX inputs)


COL1
R41
100k
COL2
R42
100k
COL3
R43
100k
COL4
R44
100k
COL5
R45
100k
COL6
R46
100k
COL7
R47
100k
COL8
R48
100k
COL9
R49
100k
COL10
R50
100k
COL11
R51
100k
COL12
R52
100k
COL13
R53
100k
COL14
R54
100k
GND

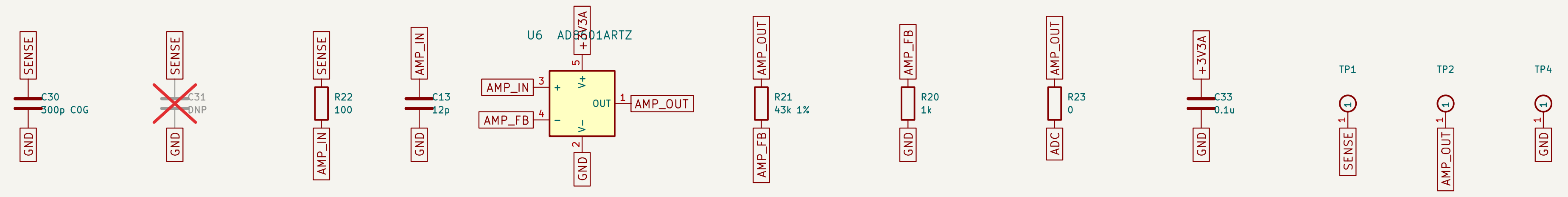
Row drive: module pads 48 / 46 / 41 / 44 (P0.24, P0.22, P0.17, P0.20) -> 100R -> ROW1..4


ROW1_DRV
R31
100
ROW1
ROW2_DRV
R32
100
ROW2
ROW3_DRV
R33
100
ROW3
ROW4_DRV
R34
100
ROW4
TP3
1
ROW1

Column select: each MUX has its own lines. INHA / INHB low = that MUX on (never both low). S1x/S2x/S3x = address A/B/C


U7 CD4051BM96
+3V3M
16
VDD
S1A
S2A
S3A
A
B
C
INHA
INH
SENSE
X
VEE
VSS
GND
X0
X1
X2
X3
X4
X5
X6
X7
COL4
COL5
COL6
GND
COL1
GND
COL2
COL3
C34
0.1u
U8 CD4051BM96
S1B
S2B
S3B
INHB
COL12
COL13
COL14
COL11
COL7
COL10
COL8
COL9
C35
0.1u

Analog front end (wired TC52 circuit): SENSE node (C30 reference cap, purge by P1.02) -> R22 -> non-inverting x44 -> AIN4 (P0.28).
C13 (12 pF at the + input, with R22: 130 MHz low-pass) keeps 2.4 GHz from the antenna out of the op amp input


SENSE
C30
500p C0G
GND
C31
DNP
R22
100
AMP_IN
C13
12p
U6 AD8601ARTZ
+3V3A
V+
V-
AMP_IN
AMP_FB
OUT
AMP_OUT
AMP_OUT
R21
43k 1%
AMP_FB
R20
1k
R23
0
ADC
C33
0.1u
TP1
TP2
TP4

+3V3A (op amp) = GPIO AFE_EN (P0.13, high drive, held high while scanning) through R27 22R, 4.7u + 100n.
The op amp draws under 1 mA; the pin also gates it for sleep. Reason for not feeding it from +3V3: docs/tc52w-build-log.md

+3V3M (MUX x2) = +3V3 through 22R, 4.7u

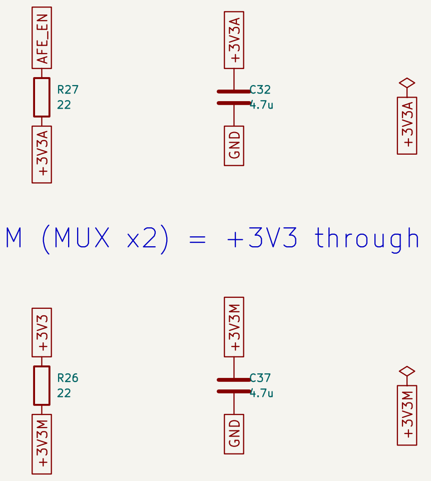
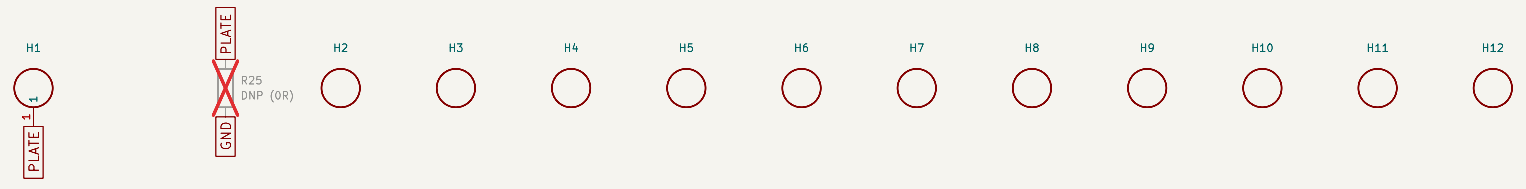

AFE_EN
R27
22
+3V3A
C32
4.7u
GND
+3V3
R26
22
+3V3M
C37
4.7u

Plate fixing holes (M2). H1 has a land: PLATE -> R25 (not fitted) -> GND


H1
PLATE
R25
DNP (0R)
GND
H2
H3
H4
H5
H6
H7
H8
H9
H10
H11
H12

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Sheet: /ec/
File: ec.kicad_sch

Title: TC52 wireless: capacitive matrix and analog front end

Size: A2
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 3/3